

The surreal and surcomplex reports

Fifteen independent mathematical research packages,
on the surreal field, its algebraic closure,
and the foundations and computation of both

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Scope

This document catalogues the fifteen research packages stored under `docs`. Each entry names the problem the report attacks, the result it claims, and the archive it arrived in.

The collection divides in three, and so does the directory tree. Six reports, under `docs/surreal`, are about the surreal field **No** itself: birthdays and their behaviour under multiplication, the option graphs of surreal forms, genetic functions and gaps, and two questions of Lipparini about evaluating formal sums. Seven, under `docs/surcomplex`, are about the algebraic closure $\mathbf{No}[i]$ and the theory of holomorphic functions over it. The last two, under `docs/foundations-and-computation`, stand outside both: they are about the subject rather than inside it, asking in what foundation this mathematics legally lives and how one computes with it exactly.

These reports were previously part of a larger collection, in a repository whose subject was elsewhere; they were moved here with their git history. That is why an entry may refer to a merge, a duplicate sweep or a delivery number: those describe how the report reached its present form, and the record is kept rather than rewritten.

Each directory holds a typeset article with its L^AT_EX source, a `README.md`, and in most cases verification code together with the recorded output of running it. A merged report additionally keeps every source manuscript verbatim under `sources/`, so the merge can be audited against what went into it. These are AI-assisted drafts. None is refereed or machine-checked, and this catalogue records what each report claims rather than verifying it.

Rebuild this document with

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latexmk -pdf manifest.tex && latexmk -c manifest.tex
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1 The surreal field

Six reports about **No**. Two questions of Lipparini on evaluating formal sums; two independent halves of Gonshor’s product-birthday conjecture, on domains neither of which contains the other; the subgraph question from Roughan’s paper on birthday arithmetic; and a genetic counterexample separating pointwise derivatives from differences of constants. The first and the fourth each arrived as a pair of archives and were merged.

1. A square-root obstruction to evaluating increasing Hahn series at ω

[surreal/hahn-evaluation-at-omega/article.pdf](#)

from `hahn_evaluation_research.zip` + `reversed_hahn_series.zip`

Answers Problem 7.7 (p. 38) of Lipparini (arXiv:2505.00424v2) negatively: there is no ring homomorphism from the field of set-supported real-coefficient Hahn series with strictly increasing surreal exponents into the surreals sending x^a to ω^a . The certificate is $B \in \mathbb{Q}[[x]]$ with $B^2 = 1 - x$, built by the recursion $b_n = -\frac{1}{2} \sum_{k=1}^{n-1} b_k b_{n-k}$: any such map would force a square to equal $1 - \omega < 0$. The homomorphism is applied only to the finite identity, so no continuity, valuation, order preservation or preservation of infinite sums is assumed. Adds the exact classification of admissible images of the variable as the infinitesimals, the monomial classification over an arbitrary ordered abelian group, and order-uniqueness on the source field. *Merged; the core certificate was*

identical in both originals and is proved once, but the two toolkits are kept — in particular a nonnegative-coefficient semiring obstruction with strictly weaker hypotheses that the square-root certificate cannot reach.

2. Gonshor’s Product-Birthday Conjecture for ordinal-support series

[surreal/gonshor-product-birthdays/surreal_product_birthdays.pdf](#)

from [gonshor_product_birthdays_research.zip](#)

Proves Gonshor’s bound $b(xy) \leq b(x) \otimes b(y)$ on the ring of Conway normal forms $\sum_{\alpha \in S} c_\alpha \omega^{-\alpha}$ with S an arbitrary set of ordinals, so transfinite and uncountable supports are allowed. Supplies an exact birthday formula by transfinite induction over the support with maintained prefix bounds and an explicit limit stage, and a complete classification of the equality cases. The route is a leading-Cantor-degree gap, which also gives that the degree of the birthday never rises under multiplication. The conjecture for arbitrary surreal numbers is not solved. The domain is incomparable with that of the Laurent-series report that follows: it excludes ω and every positive power.

3. A Laurent-series case of Gonshor’s birthday conjecture

[surreal/gonshor-laurent-birthdays/article.pdf](#)

from [Gonshor_Laurent_Birthday_Research.zip](#)

Proves the same bound $b(xy) \leq b(x) \otimes b(y)$ on the canonically embedded field $\mathbb{R}((\omega^{-1}))$, which unlike the ring of the preceding entry contains ω and its positive powers but restricts supports to order type at most ω on the integer lattice. The route is a bilinear budget: an exact splitting $b(x) = b(P) \oplus b(u)$ into polynomial and infinitesimal parts, three separate product estimates for the resulting cross terms, and a finite-jet stabilisation carrying the finite bound to infinite series without any continuity claim. Also treats finite normal forms with arbitrary ordinal exponents, derives exact birthdays for finite and infinite Laurent series, and proves a reciprocal corollary. Neither this theorem nor the preceding one contains the other; on their common corner $\mathbb{R}[[\omega^{-1}]]$ the two agree and are proved twice.

4. Canonical forms need not be subgraphs

[surreal/canonical-forms-need-not-be-subgraphs/surreal_graphs.pdf](#)

from [surreal_graphs_research.zip](#) + [surreal_sparse_graphs.zip](#)

Answers negatively the question in the Figure 3 caption of Roughan, *Surreal Birthdays and Their Arithmetic* (arXiv:1810.10373v2, p. 7): the canonical form of a surreal value need not be a subgraph of every other form of that value. Five vertices and rank three are both least possible, and there are exactly sixteen five-vertex counterexamples, eight triangle-free, four of value $1/2$. Adds cycle representations of $1/2$ at every admissible length, the exact extremal function, a planar construction of maximum degree three and arbitrarily large girth, classifications of which canonical graphs are unavoidable, and transfinite witnesses. *Merged; the two originals exhibited genuinely different five-vertex witnesses — one passing through a vertex of value -1 , one with no negative value at all — and both are printed in full, as are two odd-cycle families, two high-girth constructions and two proofs of the Fibonacci bound.*

5. A countable genetic cut across the finite–infinite gap

[surreal/genetic-gaps-and-primitives/article.pdf](#)

from [genetic_gap_counterexamples.zip](#)

With $q(t) = t/(1+t^2)$ and $B(x) = 1 - \{q(n-x) : n \in \mathbb{N} \mid \}$ as a Conway cut, proves that B is the indicator of the positive infinite surreals, is genetic under the set-sized option constructors of the cited sources, and has pointwise derivative zero everywhere. Consequently x and $x - B(x)$ are genetic primitives of 1 agreeing at every real input that do not differ by a constant, the latter also an increasing bijection. The sublevel class $\{x : 0 < x < \omega, B(x) \leq \frac{1}{2}\}$ is nonempty and bounded yet has no surreal supremum, so neither end gap supplies one.

6. A Number-Valued Broadcast Sum of Surreal Sequences

[surreal/broadcast-sum-of-surreal-sequences/article.pdf](#)

from surreal_broadcast_sum.zip

Analyses the sign-truncation game of Lipparini (arXiv:2505.00424v2, Remark 7.6(3), in connection with Problem 7.5). Proves the game always has a surreal-number value, that it agrees with finite surreal sums and with Lipparini’s ordinal operation on their respective domains, and that it fails weak coordinatewise monotonicity, with an explicit family $s[n] = 3 \cdot 2^{-n-d-1}$ on a subset E and 2^{-n-d} off it realising the failure. The underlying rule is Lipparini’s; the name and the notation are local to the report.

2 The surcomplex plane

Seven reports about $\mathbf{No}[i] = \mathbf{No}[i]$ and holomorphic function theory over it. All seven rest on one framework and all seven are merges: twenty-seven manuscripts in total, delivered on a single day as independent attempts at one subject, reduced here to seven documents.

*Read **analysis** first.* It is the foundation the other six continue, and it is where the shared calculus is established: there is no topological convergence anywhere in this theory. Every *set* of surcomplex numbers is closed and discrete, a convergent set-indexed net is eventually constant, and every continuous map $[0, 1] \rightarrow \mathbf{No}[i]$ is constant, so $\sum t^n = (1 - t)^{-1}$ is a Hahn identity and never a limit of partial sums, and contour integration is defined coefficientwise on ordinary complex paths.

*Then read the ring table in **analytic-geometry**, before anything else.* Four coefficient rings circulate through this material under nearly identical notation — the radius-free $\mathbb{C}\{z\}((t^\Gamma))$, the fixed-polydisk $\mathcal{O}(D)((t^\Gamma))$, the common-domain germ ring, and the formal $\mathbb{C}[[z]]((t^\Gamma))$ — and the inclusions between them are strict, with $F(z) = \sum_{r \geq 1} t^{rn}/(1 - rz)$ the witness: its coefficient at rn has a pole at $1/r$, so no ordinary disk carries the whole family. A theorem true over one of these rings can be false over another. Every report therefore names the ring of every statement, and records that no theorem was transported between rings; the one place a bridge is used, it is one-directional and flagged in the proof that uses it. The remaining five reports can be read in any order.

7. Surcomplex analysis: holomorphic functions on $\mathbf{No}[i]$

[surcomplex/analysis/article.pdf](#)

from surcomplex_analysis.zip and eight companion archives

One article assembled from nine manuscripts, all delivered under the same name on 21 September 2026 as independent attempts at one subject: a theory of holomorphic functions over $\mathbf{K} = \mathbf{No}[i]$ built on Hahn-supported series. On the convergence-free base described above the article develops germs, the coherent algebra $\mathcal{H}(U)$ over an ordinary domain, Weierstrass preparation and infinitesimal zero clusters, a residue calculus, the global principles, exponentials and logarithms, and an all-scale rigidity theorem. **Nine independent runs at a foundation do not agree the way nine runs at a theorem would, and the merge turned on that.** Of seventeen load-bearing notions, six are genuinely inequivalent across the sources, and a theorem proved under one is false under another: there are three inequivalent function classes (the identity theorem, the global maximum principle, Liouville and the open mapping theorem hold for coherent sections and fail for germs; sharp Schwarz and Schwarz–Pick hold for lifts and fail for coherent sections), and three inequivalent residues, two of which disagree numerically — for $1/(\zeta - \varepsilon)$ with ε infinitesimal the cluster residue at the ordinary centre 0 is 1 while the local residue at 0 is 0, the actual pole sitting at ε . A single ordinary bound in Liouville gives a characterization and not constancy: tz is real-bounded by 1 on the whole finite plane and is not constant. Eleven further conflicts are resolved and recorded, including two functions both called “the” logarithm that differ by exactly $2\pi i$. All nine manuscripts and all nine verification programs are kept; the deduplication is in the article. Its MERGE_NOTES.md is worth reading before the article.

8. Infinitesimal analytic geometry over the surcomplex numbers

[surcomplex/analytic-geometry/article.pdf](#)

from Surcomplex_Analytic_Geometry.zip + surcomplex_finite_geometry.zip

The ring theory of the infinitesimal neighbourhood — the monad $\mathfrak{m}_K^n$ of an ordinary point — rather than the deformation theory of the reports that follow. Proves, *separately of each of* the radius-free $A_n = \mathbb{C}\{z_1, \dots, z_n\}((t^\Gamma))$ and the common-domain R_n , that it is a Noetherian Jacobson domain of dimension n whose maximal ideals are exactly the infinitesimal evaluation ideals $\mathfrak{m}_a = (z_1 - a_1, \dots, z_n - a_n)$ with residue field K , regular local at each, with completion $K[[X_1, \dots, X_n]]$ — and that neither is local, each recording every infinitesimal point at once. Carries Weierstrass division and preparation, a weak and a strong Nullstellensatz, Noether normalization, finite projection, contour residue duality, Hensel–Rouché, and workspace invariance: every common zero over any larger divisible workspace already lies in K^n with unchanged multiplicity, so the count accounts for all zeros in the whole proper class $\mathbf{No}[i]$. The nonzero-value-group hypothesis is essential and not aesthetic — for $\Gamma = \{0\}$ the strong Nullstellensatz fails at $I = (0)$. *This is the report that fixes the four-ring vocabulary and proves the inclusions strict; statements holding over both A_n and R_n are stated twice with both proofs, since the two proofs use different resources and neither transfers.*

9. Finite Hahn deformations: division, conservation of multiplicity, residue duality

[surcomplex/finite-deformations/article.pdf](#)

from surcomplex_finite_deformations.zip, _finite_fibres, _finite_maps, _intersections, _research

Five manuscripts on one theorem: an ordinary holomorphic complete intersection f on a fixed polydisk, with isolated zero at the origin and colength μ , keeps its entire finite geometry under any Hahn perturbation $F = f + E$ of well-ordered *strictly positive* error support — on the unshrunk domain, with an explicit support certificate $\text{supp} \subseteq \text{supp}(G) + M$, M the monoid generated by the error support. The remainder of division is unique; the witnesses are explicitly not, their ambiguity being a Koszul syzygy carried by the same certificate. Gives the integral structure $H_\Gamma^+(D)/(F) \cong (K^\circ)^\mu$ over a valuation ring that may have high-rank value group and need not be Noetherian, with nothing completed; conservation of multiplicity at the *actual* displaced surcomplex zeros rather than their standard parts, $\sum_\zeta \mu_\zeta = \mu$; and a coefficientwise residue Res_F — defined by a series, not by a surcomplex contour integral — whose Gram determinant is a unit even at a collision and which satisfies $\text{Res}_F(G \cdot J_F) = \text{Tr}(M_G)$, individual summands routinely infinite while the total is finite. Also Hartogs, characters and local algebras, spectral invariants, trace-Jacobian and Bézoutian, monodromy and stability. The engine throughout is positive-support Neumann inversion; no argument uses convergence of partial sums in any topology.

10. Jordan separation, Cauchy theory, and Stokes calculus on the surcomplex plane

[surcomplex/contours-and-stokes/article.pdf](#)

from surcomplex_contours.zip and two same-named archives

Three manuscripts whose titles differ by one word — Stokes *Calculus, Integration, Formulae* — over one body of work. All three refuse to transfer the Jordan–Cauchy–Stokes package to $\mathbf{No}[i]$ as a unit, and all three split it the same way: a coarse Jordan separation theorem in the standard-part topology whose separator is the whole boundary monad rather than a curve; an imported semialgebraic Jordan theorem; a Hahn-coefficient de Rham complex with coefficientwise Stokes, extended to infinitesimally deformed chains; and the holomorphic consequences — Cauchy, endpoint displacement, integer winding, and microscopic circle and torus realizations of the perturbation residue — with an algebraic rational contour pairing characterized uniquely by vanishing on exact differentials and normalizing $dz/(z - a)$ to $2\pi i \cdot \text{wind}_{\text{sa}}$. Adds a purely algebraic third chain category over any characteristic-zero field, a support-preserving Poincaré lemma, an order-valued ML estimate, and a root-of-unity/Schirelman section. This is where the question the deformation report leaves open is answered: supplying an actual surcomplex contour that represents the residue series. *Two thirds of each manuscript was the same mathematics in the same order; 82 numbered results became 59, by duplicate elimination only.*

11. Global support obstructions: moving divisors, Mittag–Leffler, and a Picard dichotomy

[surcomplex/global-divisors/article.pdf](#)

from `surcomplex_global_support.zip` + `surcomplex_global_theorems.zip`

The local-to-global theory: when do infinitely many individually legitimate infinitesimal prescriptions admit one coherent global realization over the ordinary complex plane? The sharp answer for divisors is a *symmetric-support* criterion — a cluster family $P_c(u) = u^{m_c} + \sum_{j < m_c} a_{c,j} u^j$ over a closed discrete ordinary $A \subset \mathbb{C}$ is the exact zero divisor of a nonzero global section iff $\bigcup_c \bigcup_j \text{supp}(a_{c,j})$ is well ordered: the supports of the symmetric coefficients, not of separately chosen roots. Hence the two discriminating examples — $n + t^{1/n}$ alone is not realizable, its descending symmetric support $\{1/n\}$ failing, while *all* roots of $(z - n)^n = t$ are, symmetric support the singleton $\{1\}$ — and hence a globally principal effective divisor with a nonprincipal effective sub-divisor. Also Mittag–Leffler at fixed centres and for moving poles, a support-restricted Chinese remainder theory with an explicit Neumann right inverse, an exact Cousin criterion in terms of singular support, and the Picard dichotomy: every ordinary-base stalk is a nonlocal principal ideal domain, $\text{Pic}(\mathbb{C}, \mathcal{H}_\Gamma) = 0$ iff $\Gamma = \{0\}$ or $\Gamma = \mathbb{Z}\delta$, and otherwise $\dim_{\mathbb{C}} \text{Pic} \geq 2^{\aleph_0}$, with equality for countable noncyclic Γ .

12. Surcomplex polynomial algebra: root geometry, Newton profiles, residue duality

[surcomplex/polynomial-algebra/article.pdf](#)

from `surcomplex_polynomial_algebra.zip` and two same-named archives

Finite-degree polynomial algebra over $\mathbf{No}[i]$, worked inside set-sized algebraically closed Hahn workspaces and keeping two geometries deliberately apart throughout: the surreal-valued modulus, whose disks are ordered, and the natural Hahn valuation, whose balls have arbitrary ordered-group rank. A Rouché theorem in each. Covers the finite polynomial algebra and the fundamental theorem there, CRT/Hermite jets, resultants and discriminants, ordered root geometry (root bounds, Gauss–Lucas, Jensen disks, a sharp Schoenberg second-moment inequality with its equality case), real-closed-field transfer with Hermite signature counting, exact root loci under coefficient uncertainty in both geometries, Newton profiles over an arbitrary ordered exponent group, image balls and branch values, exact Rolle counts and the finite root-cluster tree, support-controlled coprime Hensel factorization with a fixed-domain parameter form that shrinks no neighbourhood, and universal residue duality with the trace identity. *Three perturbation theorems with three different hypotheses are kept separate rather than collapsed, as are two discriminant layer formulas and two mapping theorems, with their agreement proved; the alternative was to lose what each source reached that the others did not.*

13. Trigonometry on the surcomplex plane

[surcomplex/trigonometry/article.pdf](#)

from `surcomplex_trigonometry.zip` and two same-named archives

Its organizing theorem is that every surcomplex direction has a *finite* surreal angle: $\text{cis} : (\mathcal{O}_{\mathbb{R}}, +) \rightarrow (T(\mathbf{No}), \cdot)$ is surjective with kernel exactly $2\pi\mathbb{Z}$, so $\mathcal{O}_{\mathbb{R}}/2\pi\mathbb{Z} \cong T(\mathbf{No}) \cong T(\mathbb{R}) \times (\mathfrak{m}_{\mathbb{R}}, +)$, and the whole of plane, spherical and hyperbolic triangle geometry is available at arbitrary surreal length scale without ever choosing a value for $\sin \omega$. From it: canonical sine and cosine by Hahn summation, the unit circle with its branch conventions and projective half-angle chart, inverse functions on their full surreal domains, the angle-sum, cosine, sine, tangent, half-angle, Heron, incircle, Euler, angle-bisector, cevian, Ceva, inscribed-angle and Ptolemy theorems, an exact valuation dictionary between sides, angles and area with a quadratic triangle-inequality defect formula and two flat-triangle theorems, a rank-two collision-stable intersection algebra, the canonical complex extension to the strip where sine is surjective with completely determined fibres, finite Fourier inversion, Parseval and a surcomplex Fejér–Riesz factorization, and the classification of all global phase extensions, with the Ehrlich–Kaplan normalization placed inside that classification. Two coherence obstructions close it: there is no coherent bounded sine of infinite frequency, and all-scale coherence forces a polynomial. *No mathematical disagreement*

was found between the three sources; the duplication was near-total on the first two thirds and the divergence near-total on the last.

3 Foundations and computation

Two reports that take the rest of the collection as their object. Neither belongs under **surreal** or **surcomplex**, because both range over the whole subject. Each is a merge of three manuscripts delivered together on 21 September 2026, and in each the hazard was not overlap but a collision of near-identical names.

14. Foundations for surreal and surcomplex mathematics

[foundations-and-computation/foundations/article.pdf](#)

from `surreal_surcomplex_foundations.zip` + `surcomplex_foundations.zip` + `surreal_foundations.zip`

In what foundation does this mathematics legally live, and how would a proof assistant encode it? Compares ZF/ZFC with definable classes, GB/GBC/NBG, Kelley–Morse, Grothendieck universes and TG, Feferman reflection, constructive set theory, dependent type theory, univalent foundations and native Lean, and proves the size obstructions each must respect: no unrestricted cut operation can accept every subclass of its own carrier, the surreals are not Dedekind complete, and a universe cannot cut itself. The positive counterpart is a localization theorem — every *set* of surcomplex numbers already lies in one divisible set-sized workspace $K_\Gamma = \mathbb{C}((t^\Gamma))$ — with a finite-algebra descent theorem explaining why a finite zero scheme gains no points in a larger workspace. Carries two series over $\mathbb{C}((t^\mathbb{Q}))$ that differ only in the sign of an exponent and reach opposite conclusions: $\sum t^{-n^2} X^n$ is summable at no nonzero argument, while $\sum t^{+n^2} z^n$ is summable everywhere and is a *nonpolynomial* function coherent at every radius — which does not refute the all-scale rigidity theorem of **surcomplex/analysis**, stated for class functions on the whole **No**[*i*], but shows it does not transfer to a fixed value group. *Merged; the danger was aggregation. All three manuscripts read the same Lean repository at the same revision on the same day, so their agreement is one act of source reading performed three times, not three confirmations. The findings are printed once and the verification boundary has its own early section.* Three of the five shipped Lean files were compiled for this repository and `#print axioms` run on every declaration: no `sorryAx` and no `Classical.choice` anywhere, and both size-obstruction theorems depend on no axioms at all. The toolchain used was not the one the sources pin, and one shipped file does not compile as delivered.

15. Surreal and surcomplex numbers in symbolic computer algebra

[foundations-and-computation/computer-algebra/article.pdf](#)

from `surreal_surcomplex_cas.zip` and two same-named archives

What can actually be computed exactly, and what only approximated or denoted? Separates finite exact denotation, effective coefficient access, decidable equality and order, and certified approximation into capability tiers, then works through an ordered rational-monomial core, real-algebraic closure, certified Hahn series, formal germs, analytic lifts, transseries, Newton profiles, Hensel lifting, finite quotient algebras, residues, and a proposed Wolfram Language architecture. *Merged; the danger was three prototypes with one name.* Each manuscript ships an executable Wolfram package, two of them near-anagrams of each other, two exporting six identical public symbol names with incompatible signatures, and all three calling themselves the rational monomial core. They are three different objects: coefficients $\mathbb{Q}$ in one and $\mathbb{Q}(i)$ in the others; exponent lattices $\mathbb{Z}^d$ against $\mathbb{Q}^r$, so a ramified root is representable in one and not the next; and two of them order their monomial variables by opposite conventions, which reverses every inequality. The report never writes “the prototype” and tabulates the divergences instead. All three check suites were rerun for this repository and reproduce exactly — 34/34, 110/110 and 88/88 in Wolfram, 69/69 and 323/323 in Python — but one package cannot be loaded at all as shipped: an unclosed bracket stops it parsing, which is why its own record submits definitions rather than loading the file.